
Shibboleth: Probe-and-Replay Model-Response Watermarking for Diffusion Language Models

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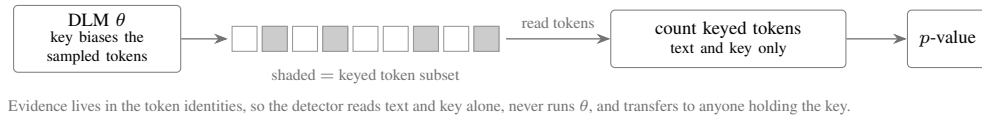
Abstract

1 We propose Shibboleth, a model-response watermarking method for masked dif-
2 fusion language models (DLMs). Unlike an autoregressive sampler, a masked
3 DLM can be re-evaluated on its own output under an arbitrary remasking of the
4 context. Because a practical DLM, unlike an ideal predictor, shifts the likelihood
5 of an emitted token by an amount its own weights determine, a watermarking
6 channel opens in the relation between a text and the model rather than in the tokens.
7 Shibboleth evaluates a keyed family of context probes against each masked block,
8 admits as carriers only the unresolved positions whose block-masked conditional
9 retains mass on both response signs, and resamples at those carriers from the live
10 conditional until the requested sign is committed. Unlike logit-bias methods it
11 leaves the model’s probabilities untouched and conditions only on which model-
12 drawn proposal is accepted, and unlike order-steering methods it spends none of
13 the decoding schedule. Because the admitting gate is invariant to swapping two
14 probes, it stays blind to the sign the key will request, and carriers may therefore be
15 discovered adaptively without disturbing the null. Detection re-masks the finished
16 text, rebuilds the identical probes and carriers, and counts keyed sign matches,
17 whose distribution on text written without the key is exactly binomial, and p -values
18 are therefore exact at finite sample size and under carriers the method itself chose,
19 which no threshold fitted to a corpus can promise. On LLaDA-8B-Instruct it
20 reaches 0.80 TPR at 1% FPR, below the strongest token-based detector but at the
21 only perplexity ratio in the comparison that reads below its own unwatermarked
22 baseline. Verification in exchange costs one rerun of the checkpoint, and evidence
23 written against a prefix does not survive editing, leaving the method suited to a
24 verifier that holds the checkpoint rather than to open screening.

25 1 Introduction

26 Language models now produce text that readers cannot reliably separate from human writing, which
27 turns provenance into a security problem, since disinformation, phishing, and academic fraud all
28 grow cheaper once authorship can no longer be established [23]. Institutions facing that problem
29 reach first for a post-hoc detector, and academic-integrity screening is the clearest case, given that an
30 essay’s authorship has to be settled after it is submitted [19]. A detector of that kind is nevertheless
31 fitted to one generation process, and its verdicts weaken as deployment moves to models that decode
32 by a different mechanism, masked diffusion models among them. Watermarking is the answer most
33 widely deployed against it, since it plants a keyed statistical signal during generation and lets whoever
34 holds the key test a later document for it [17, 27]. A watermark is nevertheless worth only the
35 guarantee attached to its verdict, given that a false positive is an accusation against a human author,
36 and a deployment therefore needs a false-positive rate it can state in advance rather than one fitted
37 to a calibration corpus. Since adversaries also paraphrase output and learn a key’s behaviour from

Token-based watermark



Shibboleth (this work)

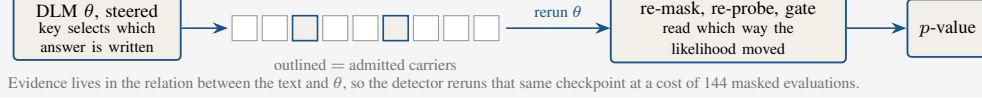


Figure 1: **Two places to keep a watermark.** A token-based watermark biases which tokens are sampled and recovers that bias by reading the text against the key. Shibboleth instead writes the answer the key requests to a keyed probe of the context, which makes the evidence a property of the text together with one checkpoint and obliges the detector to rerun it.

38 detector queries [15, 19], the guarantees worth designing for are the ones that survive finite samples
 39 and adaptively chosen trials [9, 38].

40 Existing watermarks place their secret in the tokens, whether by biasing keyed vocabulary subsets [17],
 41 by coupling each draw to pseudorandomness [20], or, in recent diffusion language model (DLM)
 42 methods, by exploiting unresolved context and denoising order [14]. Detection then reduces to
 43 inspecting the text together with the key, and the generator is never consulted. Masked DLMs,
 44 by contrast, permit a different arrangement, since a finished text can be fed back to the model
 45 with selected context hidden and the likelihood of an emitted token then moves by an amount the
 46 checkpoint alone determines. Hiding a keyed subset of the context is in this sense a *context probe*, the
 47 direction in which the likelihood moves is the model’s response to it, and provenance can be argued
 48 from the agreement between the two. Such a probe is worth applying only where the model has a
 49 genuine choice of answer, that is where several plausible tokens coexist at a masked position and
 50 react with opposite signs to two patterns.

51 Moving the secret out of the tokens and into that relation, however, costs two things a token-local
 52 detector obtains for nothing, since the verifier must apply exactly the probe the embedder applied,
 53 and the choice of carriers must be settled before the accepted answer is known; a gate with sight of
 54 the rewarded sign would otherwise inflate the false-positive rate above the level the detector reports.
 55 Section 3.2 states both as requirements, and existing DLM watermarks avoid them by keeping
 56 detection token-local, whether by removing prefix dependence [8], by averaging a red–green bias
 57 over unresolved context [12], or by steering which mask is resolved next [14], none of which asks the
 58 model anything about the text it produced.

59 Shibboleth instead meets each condition with one mechanism (Figure 1). It settles replayability by
 60 re-masking the current block together with its suffix before scoring the probe family, and the state it
 61 measures therefore follows from the final text alone. A *balanced gate* then admits as carriers only the
 62 unresolved positions whose block-masked conditional retains mass on both response signs, scoring
 63 them in a way that is invariant to swapping the two patterns and therefore blind to the sign the key
 64 will ask for. Writing the answer is the easier half, since Shibboleth draws at most R proposals from
 65 the unchanged live row at an admitted position and commits the first one carrying that sign, which
 66 amounts to rejection sampling and does change the output distribution, although every committed
 67 token remains one the model proposed. Verification afterwards re-applies the same probe, reapplies
 68 the gate, and returns an exact binomial tail. Unlike a prefix-based watermark, which can require strict
 69 left-to-right decoding, and unlike an order-guided one, which changes which mask is resolved next,
 70 Shibboleth acts only after the reference schedule has produced a row, and therefore leaves the block
 71 partition and the confidence order untouched.

- 72 • We show across 36 settings that this checkpoint’s conditionals are too sharp for answers to be
 73 inserted after the fact, and the answer must therefore be written while the conditional is still live
 74 (Section 3.2).
- 75 • We propose Shibboleth, in which block-and-suffix re-masking makes the probe replayable from
 76 the finished text and orientation-blind gating keeps the accepted answer independent of how the

77 carriers were discovered, which together keep the detector’s p -values exact and free of calibration
78 (Sections 3.3 to 3.5).

- 79 • We evaluate Shibboleth on LLaDA-8B-Instruct and report both sides of the trade, since the
80 strongest setting reaches 0.80 TPR at 1% FPR at the best perplexity ratio of any method compared,
81 whereas verification spends 144 masked evaluations and TPR collapses under 10% re-denoising
82 (Section 4).

83 2 Related Work

84 **Diffusion language models.** Discrete and masked diffusion objectives carry diffusion [13] over to
85 text [3, 24, 26, 31, 32] and now support instruction-following systems such as LLaDA and Dream
86 [25, 36, 39], with commercial deployments following the same recipe [21]. Because the order in
87 which positions commit affects quality and throughput [2, 6, 16, 33, 34], and because a run can equally
88 be stopped before its step budget is spent [22], accelerators treat the whole schedule as a tunable
89 degree of freedom. Shibboleth, by contrast, treats it as an invariant and draws on masked-token
90 reconstruction only to define a probe it can later repeat.

91 **Watermarking autoregressive text.** Autoregressive watermarks bias keyed token subsets [17] or
92 couple sampling to keyed randomness [1, 11, 20], with later work characterising their reliability
93 and undetectability [9, 18, 38]. Detectors of this family are inexpensive precisely because a key-
94 token hash needs no generator, yet that hash is defined over a resolved prefix and does not transfer
95 unchanged to a context in which most token positions are still masked.

96 **Watermarking diffusion language models.** DLM-specific methods restore a well-defined keyed
97 partition by other means, whether by removing prefix dependence [8], by averaging a red-green
98 bias over unresolved context [12], by keyed sampling [4], or by exploiting the decoding order
99 itself [14, 28, 35]. We compare against dgMARK, the strongest order-steering baseline with a
100 zero-inference detector. Shibboleth differs from all of them in where the evidence resides, since
101 dgMARK spends the DLM’s freedom over commit order and reads the outcome off the text, whereas
102 Shibboleth spends nothing from the schedule and instead has the verifier rerun the exact checkpoint,
103 buying a statistic bound to model behaviour at the price of a detector that no longer runs on text alone.

104 **Detection that consults the model.** Verifiers needing more than the text are not new, since
105 GaussMark places its key in a Gaussian perturbation of the model’s own weights [7], and wa-
106 termarks built to expose model theft assume the verifier can query the suspect system [37]. Unlike
107 GaussMark, which ships a perturbed checkpoint, Shibboleth leaves the released weights as they are
108 and puts the key in the questions asked of them, and one deployed model therefore serves every
109 document key.

110 3 Shibboleth: Watermarking by Model Response

111 3.1 Decoding and Notation

112 Masked diffusion language models differ from autoregressive models in that they predict a missing
113 token from an arbitrary subset of revealed tokens, and the same sequence can consequently be
114 produced under many decoding orders. Let x be a prompt and $y = (y_1, \dots, y_N)$ the continuation to
115 be generated, and for a revealed set $I \subset \{1, \dots, N\}$ with a target position $i \notin I$, the model supplies
116 a predictor $p_\theta(y_i \mid y_I, x)$ while the remaining positions carry the mask symbol, so that one sequence
117 evaluation returns that distribution at every unresolved position at once.

118 Practical decoders nevertheless leave much of that freedom unused, and we study LLaDA’s semi-
119 autoregressive schedule, in which the continuation is partitioned into blocks that finish from left to
120 right while diffusion resolves the masks inside the active block [2, 25], and the reference confidence
121 rule decides which positions commit at each step. Everything below is therefore stated for one block
122 B_b with the prefix F_b that precedes it.

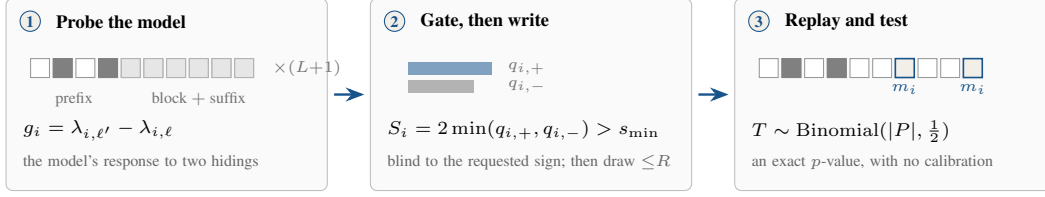


Figure 2: **How Shibboleth writes a mark and reads it back.** The same three steps run at generation and at verification, which is what makes the evidence replayable, and the gate in the middle step is settled before the key names the sign it accepts, which is what keeps its p -values exact.

123 3.2 Problem Setup

124 Our goal is a watermark whose detector consults the model rather than the text alone, telling a
 125 continuation generated under a secret key from one generated without it at a false-positive rate the
 126 verifier states in advance, while disturbing the decoding schedule and the output quality as little as
 127 the channel allows.

128 **Embedder and verifier.** The embedder receives the model, the prompt, a secret key K , and the
 129 target bits, whereas the verifier later receives only the completed sequence, the same model, K , and
 130 the public parameters, without logits, random seeds, or commit order. Each document then draws
 131 its own key from K and a fresh nonce through a keyed hash, $K_d = \text{HMAC}(K, \text{nonce}_d)$, which to
 132 anyone without K behaves as a random function of the nonce [5]. That document key generates both
 133 the probe family and the accepted answers, and the false-positive guarantee covers any text produced
 134 independently of it.

135 **Threat model.** Both the model and the algorithm may be public, since secrecy rests entirely in
 136 K . Concretely, the detector does not accuse innocent text, meaning that for every text written
 137 independently of the document key, machine text from another sampler included, a detector run
 138 at nominal level α fires with probability at most α , and the bound holds at finite sample size and
 139 under carriers the method itself selects. We claim nothing in the complementary direction, and an
 140 adversary willing to edit the output can therefore suppress detection, as Section 4.2 measures. We
 141 likewise claim nothing after key disclosure, under the detector-query access that recovers a key's
 142 behaviour [15], or when verification runs a numerically different checkpoint.

143 **Post-hoc substitution.** An embedder that swapped tokens in the finished text would be cheaper
 144 than one intervening during sampling, and we begin by measuring how much room such a swap
 145 has. Write τ for the likelihood budget of a substitution, the number of nats by which a replacement
 146 may fall below the live argmax. Across 36 settings that replace tokens after ordinary generation, the
 147 median selected position admits exactly one token at $\tau = 3$, the mean rising only from 1.6 at $\tau = 3$
 148 to 5.6 at $\tau = 6$, and the best setting inside a $1.35 \times$ perplexity budget reaches no more than 0.10 TPR
 149 at 1% FPR. Conditionals this sharp leave too little room for a forced replacement, and the embedder
 150 must therefore write while the conditional is still live.

151 We accordingly formalise the problem with three requirements.

- 152 1. **Replayability.** The verifier must be able to put exactly the question the embedder put, from the
 153 finished text and the key alone, with no generation trajectory, stored logits, or record of rejected
 154 samples.
- 155 2. **Blind selection.** The set of positions that carry a bit may depend arbitrarily on the text and
 156 the model, but that choice has to be settled before the sign the key rewards is consulted, since
 157 otherwise the reported false-positive rate understates the true one.
- 158 3. **Schedule preservation.** The block partition and the confidence order must survive unchanged,
 159 so that a production decoder keeps the latency and quality profile it was tuned for.

160 The first two pull against each other, since replayability wants every scored context recoverable from
 161 a text the model helped choose while blind selection wants that choice to say nothing about the sign
 162 the key rewards. Sections 3.3 to 3.5 meet each requirement with one mechanism.

3.3 Probing the Model and Gating the Carriers

Unlike dgMARK, which obtains its channel by choosing which mask is resolved next, Shibboleth leaves the decoding schedule as the reference decoder set it. For block B_b , let F_b be the prompt together with the completed prefix, and let θ denote the reference checkpoint with p_θ for its predictor. Shibboleth then constructs L equal-size near, far, and keyed pseudorandom *probe patterns* $D_{b,\ell} \subset F_b$ indexed by $\ell \in \{1, \dots, L\}$, each the set of prefix positions it hides, and it masks $D_{b,\ell}$ together with the current block and the suffix, and one model evaluation therefore yields, for every token position $i \in B_b$ and every vocabulary item v , the log-probability

$$\lambda_{i,\ell}(v) = \log p_\theta(v \mid F_b \setminus D_{b,\ell}; B_b \text{ and suffix masked}). \quad (1)$$

Probing one block therefore costs $L + 1$ sequence evaluations, one per pattern plus an unprobed baseline, and the verifier later repeats the same set.

Re-masking the block and the suffix removes every token unavailable when the probe was first applied, which is what buys replayability: the verifier restores the same state from the final text alone, and a transient generation-time response becomes a document statistic. Masking only the probed prefix would fail on both sides, since token positions committed later in the same block would leak into verification-time logits while the future suffix was invisible during embedding.

For an unordered pair of pattern indices (ℓ, ℓ') and a candidate token z , define the contrast $g_i^{\ell,\ell'}(z) = \lambda_{i,\ell'}(z) - \lambda_{i,\ell}(z)$. Let p_i^{base} denote the block-masked conditional, the model distribution when B_b and the suffix are masked but no probe pattern is applied. The mass on each sign and the two-sided score built from them are

$$q_{i,\pm}^{\ell,\ell'} = \sum_z p_i^{\text{base}}(z) \mathbf{1}[\pm g_i^{\ell,\ell'}(z) > 0], \quad S_i^{\ell,\ell'} = 2 \min(q_{i,+}^{\ell,\ell'}, q_{i,-}^{\ell,\ell'}). \quad (2)$$

Shibboleth selects, independently at each token position, the pair maximizing $S_i^{\ell,\ell'}$ and stores its contrast as g_i in the bank $\mathcal{R}_b$, after which the gate admits position i whenever $S_i > s_{\min}$, subject to an optional per-block cap. Because $S_i^{\ell,\ell'}$ is unchanged when ℓ and ℓ' are swapped, the gate neither observes nor favours the orientation the key will request, which is what buys blind selection, and both the pair selection and the gating recompute from the block-masked state. Best-pair selection is moreover what makes the gate productive, raising the measured mean score from 0.28 for a fixed near/far pair to 0.45, although the score stays a symmetric proxy computed on the block-masked conditional while generation runs from an evolving live one.

Once the gate has admitted a token position, the key supplies an orientation $\epsilon_i \in \{-1, +1\}$ from a PRF domain separate from pattern generation, pair selection, carrier assignment, and payload grouping. A target $w_i \in \{-1, +1\}$ then carries the desired provenance or message bit, so that for a candidate z the shared embedding-and-verification predicate becomes

$$m_i(z) = \mathbf{1}[w_i \epsilon_i g_i(z) > 0]. \quad (3)$$

Should $g_i(z)$ vanish exactly, a second domain-separated fair bit defines $m_i(z)$, and although such ties cannot be steered, the tie rule keeps the detector's p -values exact.

3.4 Writing the Requested Answer

Shibboleth inherits the reference confidence order together with the live conditional p_i , and the token positions that the gate rejected are sampled normally. At an admitted carrier define

$$A_i = \left\{ z : w_i \epsilon_i g_i(z) > 0 \text{ and } p_i(z) \geq \kappa \max_v p_i(v) \right\}. \quad (4)$$

It draws from p_i up to R times and commits the first member of A_i , and after R misses it commits one fresh unchecked draw instead. Because the row is never recomputed, those retries cost no model evaluation.

That procedure changes the sampling distribution, and by a stated amount. Let $a_i = p_i(A_i)$ be the acceptance mass and let $q_i = \sum_z p_i(z) m_i(z)$, where $m_i(z)$ applies Eq. (3) and the tie rule to candidate z , so that q_i is the probability that an unconditional draw scores as a detector hit irrespective of the plausibility floor. The committed-token distribution is

$$(1 - (1 - a_i)^R) p_i(\cdot \mid A_i) + (1 - a_i)^R p_i, \quad (5)$$

206 and its detector-hit probability is

$$\Pr[m_i = 1] = 1 - (1 - a_i)^R(1 - q_i). \quad (6)$$

207 Whenever $\kappa = 0$ and the conditional places no mass on exact ties, $a_i = q_i$ and Eq. (6) collapses to
 208 $1 - (1 - q_i)^{R+1}$, so that a_i and q_i diverge only under a floor or a tie rule, where an unchecked fallback
 209 may score as a hit without belonging to A_i . The floor meanwhile bounds each guided proposal’s
 210 model NLL to within $-\log \kappa$ of the live argmax but constrains neither the fallback nor semantic
 211 quality, which is why being model-proposed remains a support statement rather than a distortion-free
 212 claim.

213 3.5 Watermark Detection

214 Given a completed text y , the verifier re-masks each block and suffix, rebuilds the probe from the
 215 fixed prefix, reapplies the gate, and evaluates $m_i = m_i(y_i)$ at the admitted positions, so that rejected
 216 proposals and the generation trajectory are irrelevant. For provenance-only detection, which is all
 217 we evaluate here, the admitted carriers form a presence pool P with $w_i = +1$, whereas a payload
 218 implementation would keep the group for each payload bit separate, so that a balanced message
 219 cannot cancel positive against negative targets. The presence statistic and its one-sided p -value are

$$T = \sum_{i \in P} m_i, \quad p_{\text{det}} = \Pr[\text{Binomial}(|P|, \tfrac{1}{2}) \geq T]. \quad (7)$$

220 **Proposition 1** (Exact false-positive control). *Assume the PRF domains above behave as independent*
 221 *random functions. Fix any text y written without the document key, and condition on the carrier set,*
 222 *selected pairs, contrast values, payload roles, and target bits. The orientation bits (and tie bits where*
 223 *needed) remain independent fair coins. Consequently the m_i are independent $\text{Bernoulli}(\frac{1}{2})$ variables*
 224 *and $T \sim \text{Binomial}(|P|, \frac{1}{2})$ exactly, so that p_{det} in Eq. (7) is an exact p -value and thresholding it at*
 225 *α holds the false-positive rate at or below α for every document length and carrier count.*

226 The conditioning carries the result, since carrier count, selected pair, and contrast magnitude may all
 227 depend arbitrarily on the text and the model, yet each is fixed before the independent orientation bit
 228 is consulted. Flipping ϵ_i therefore flips m_i whenever $g_i(y_i) \neq 0$, the tie rule supplies a fresh fair bit
 229 when it vanishes, and the carriers stay independent of one another throughout. In particular, neither an
 230 asymptotic approximation nor a calibration corpus enters anywhere, unlike a z -score whose reference
 231 distribution has to be estimated. The guarantee does nevertheless assume text written without the
 232 key, together with independence among the several uses of the key, which the domain-separated hash
 233 supplies.

234 4 Experiments

235 **Datasets and setup.** We evaluate GSAI-ML/LLaDA-8B-Instruct [25] in fp16 on a single TITAN
 236 RTX. Detectability is measured on C4 realnewslike [30] prompts truncated to 300 characters
 237 as dgMARK’s protocol prescribes. Shibboleth runs produce 512 tokens in 32-token blocks at one
 238 denoising step per token under the native confidence schedule with full-vocabulary multinomial
 239 sampling at temperature 0.8; the token-based baselines are local reproductions at 256 tokens under
 240 each method’s own decoding regime, and the table states both lengths rather than pretending to
 241 one protocol. On top of the reference sampler, Shibboleth adds $L = 8$ probe patterns, threshold
 242 $s_{\min} = 0.5$, retry budget R , and an optional floor κ , and we test provenance detection ($w_i = +1$)
 243 rather than payload recovery.

244 **Metrics and baselines.** Detectability is TPR at analytic FPR thresholds from Eq. (7), and quality
 245 is GPT-2-large [29] perplexity against an unwatermarked baseline drawn from the same decoder,
 246 reported as $\exp(\text{mean}(\text{NLL}_{\text{wm}} - \text{NLL}_{\text{ctrl}}))$. That ratio measures the distortion each method adds to
 247 its own unwatermarked baseline rather than absolute text quality, and comparisons of ratios across
 248 methods are therefore indicative rather than controlled. Compute counts sequence evaluations, and
 249 each of the $512/32 = 16$ blocks accordingly needs $L + 1 = 9$ of them, hence 144 for verification
 250 against $512 + 144 = 656$ for generation. We compare against KGW [17] under left-to-right decoding
 251 and dgMARK [14] under confidence decoding.

Method	Setting	PPL ratio ↓	TPR at FPR ↑		Sequence evaluations ↓	
			1%	0.1%	Generation	Detection
KGW [17]	$\delta = 1$, 256 tok	$1.03\times$	0.93	0.73	256	0
dgMARK [14]	$k = 1$, 256 tok	$1.55\times$	0.74	0.62	256	0
dgMARK [14]	+3-beam, 256 tok	$1.23\times$	0.86	0.82	~ 1024	0
Shibboleth	$R = 1$, 512 tok	$0.97\times$	0.43	0.20	656	144
Shibboleth	$R = 8, \kappa = 0.1$, 512 tok	$0.87\times$	0.70	0.60	656	144
Shibboleth	$R = 16, \kappa = 0.05$, 512 tok	$0.74\times$	0.80	0.60	656	144

Table 1: **Watermark detectability.** Detection rate, perplexity against each method’s own matched unwatermarked baseline, and compute in sequence evaluations. All rows are local reproductions on one checkpoint and prompt protocol, but Shibboleth uses 512-token outputs while the token-based baselines use 256, and each block keeps its own decoding regime, so ratios are comparable within a method and indicative across methods. A PPL ratio below one is not evidence of better semantics. Principal Shibboleth rows use $n = 30$ documents ($n = 20$ for $R = 16$); dgMARK uses $n = 50$. Here δ is KGW’s green-list logit bias and k is dgMARK’s candidate count.

4.1 Main Results and Analysis

Retry budget and the floor. Detection scales with the number of guided draws as Eq. (6) predicts, since $R = 1$ reaches 0.43 TPR at 1% FPR and raising R from 2 to 4 to 8 on 30 held-out prompts lifts it from 0.73 to 0.83 to 0.87. Retries cost no model evaluation, and the budget is therefore paid in quality instead, with unfloored perplexity ratios climbing from $1.14\times$ at $R = 2$ to $1.40\times$ at $R = 4$ and $1.61\times$ at $R = 8$. The floor converts part of that back, since $(R, \kappa) = (8, 0.1)$ obtains 0.70 TPR at $0.87\times$ and $(16, 0.05)$ obtains 0.80 at $0.74\times$, both below their unwatermarked baselines, although a ratio below one indicates only that GPT-2-large prefers the high-probability tokens the floor admits.

Downstream quality. Perplexity ratios are a proxy, so we additionally measure GSM8K [10] accuracy on 50 test problems at 256 tokens, comparing the watermarked sampler against its matched unwatermarked control under the identical decoding schedule; results are ?? pending the run.

Cost against token-local detectors. Detection trails the strongest token-based detector: KGW reaches 0.93 TPR at 1% FPR against 0.80 for Shibboleth, and dgMARK’s three-beam variant reaches 0.82 at 0.1% FPR against 0.60. Shibboleth instead carries the only perplexity ratio in the comparison that reads below its own unwatermarked baseline, $0.74\times$ against $1.03\times$ for KGW and $1.23\times$ for three-beam dgMARK; every baseline row costs perplexity, whereas the floor discards exactly the low-probability draws a proxy scorer punishes. KGW additionally requires strict left-to-right decoding, and its repeated bigram events must be deduplicated, without which its control measured 37% FPR. Shibboleth alone, against all of that, spends 144 masked evaluations to detect at all. Cohorts of twenty to fifty documents leave a standard error near 0.08 on a TPR of 0.80, and two rows separated by less than about a tenth should accordingly be read as a tie.

4.2 Calibration and Robustness

Calibration under the shipped rule. Proposition 1 is analytical, which is why we also measure the rule as deployed. Across 20 keys and 120 non-watermarked generations apiece, the mean FPR is 0.0121, 0.0037 and 0.0000 at nominal 0.10, 0.05 and 0.01, and the worst individual key gives 0.0583, 0.0250 and 0.0000, and control consequently holds for every key rather than on average alone, which is what binds when one key serves many documents.

Post-editing. Edit robustness is the channel’s principal failure mode, and it degrades abruptly, since same-model argmax re-denoising of a random 10% of token positions takes TPR at 1% FPR from 0.35 to 0.05 on 20 $R = 1$ documents and deleting 10% measures the same. The collapse is one of synchronisation rather than signal strength, because an early edit alters the fixed prefix from which every later probe is rebuilt, and a single change thus invalidates every carrier after it. That cohort uses $R = 1$, and the headline setting at $R = 16$ consequently remains unattacked, which we record as an open gap rather than as evidence of resilience.

5 Conclusion

Shibboleth argues provenance from a keyed probe that a masked DLM answers about its own output rather than from a pattern left in the tokens, and because the gate is settled before the key names the sign it accepts, carriers may be discovered adaptively at no cost to the detector’s exact p -values. That trade is worth making where the decoding schedule must not change and the evidence wanted is of the kind a specified model can vouch for, which a key–token hash cannot supply, since anyone holding the key can compute it without the generator. Absent either constraint, a token-local watermark is cheaper and easier to resynchronise after edits.

Limitations. That guarantee is nevertheless one-sided, since soundness holds for any text written without the key while nothing is claimed in the other direction, and a single same-model denoising pass over a tenth of the positions accordingly takes detection from 0.35 to 0.05 at $R = 1$, leaving Shibboleth with no claim of robustness to editing at any setting. Cost is the second boundary, since 144 masked evaluations per document rule out screening a corpus. The evidence is bound to one checkpoint as well, and a verifier holding different weights, or the same weights at a different precision, consequently cannot reproduce the responses at all. Finally, the evaluation is itself narrow, covering one masked diffusion model, one prompt corpus, cohorts of twenty to fifty documents, quality proxies read as ratios against each method’s own unwatermarked baseline, and provenance detection only, with the payload construction of Section 3.5 never measured.

Future work. Each boundary suggests its own repair. Anchoring probe patterns to content rather than to absolute positions would stop an edit from desynchronising every carrier after it, which is the single change most likely to turn an abrupt collapse into a graceful decline, and the same study should attack the strongest configuration rather than the weakest one. Verification cost is the easier target, since the binomial tail is monotone in the matches so far and a sequential test could stop once the evidence is decisive. Loosening the checkpoint requirement in turn asks for a contrast quantised coarsely enough that numerical drift cannot flip a sign, trading a little capacity for portability. Adaptive adversaries deserve their own study, since a detector answering queries leaks something about its key. Table 1 finally argues for composition rather than a winner, with a cheap token-local screen deciding what is worth a model-response check.

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